



```
In [10]: import pandas as pd
import numpy as np
import seaborn as sns
from sklearn.model_selection import train_test_split
```

```
In [13]: #Question1
dataframe=pd.DataFrame(pd.read_csv('T1_online_shoppers_intention - T1_online_s
```

```
In [14]: #Question2
print("Shape for given dataset is ",dataframe.shape)
```

Shape for given dataset is (12330, 18)

```
In [20]: #Question3
print("Datatypes for the columns in dataset are")
dataframe.dtypes
```

Datatypes for the columns in dataset are

```
Out[20]: Administrative      int64
Administrative_Duration    float64
Informational              int64
Informational_Duration     float64
ProductRelated            int64
ProductRelated_Duration   float64
BounceRates               float64
ExitRates                 float64
PageValues                float64
SpecialDay                float64
Month                     object
OperatingSystems          int64
Browser                   int64
Region                    int64
TrafficType               int64
VisitorType               object
Weekend                   bool
Revenue                   bool
dtype: object
```

```
In [21]: #Question4
print("Number of zeros in each column are")
print((dataframe==0).sum())
```

```

Number of zeros in each column are
Administrative          5768
Administrative_Duration 5903
Informational           9699
Informational_Duration  9925
ProductRelated          38
ProductRelated_Duration 755
BounceRates            5518
ExitRates              76
PageValues             9600
SpecialDay             11079
Month                  0
OperatingSystems        0
Browser                0
Region                 0
TrafficType            0
VisitorType            0
Weekend                9462
Revenue                10422
dtype: int64

```

```

In [24]: #Question5
print("Number of null values or NaN values are given as")
print((dataframe.isnull()).sum())

```

```

Number of null values or NaN values are given as
Administrative          0
Administrative_Duration 0
Informational           0
Informational_Duration  0
ProductRelated          0
ProductRelated_Duration 0
BounceRates            0
ExitRates              0
PageValues             0
SpecialDay             0
Month                  0
OperatingSystems        0
Browser                0
Region                 0
TrafficType            0
VisitorType            0
Weekend                0
Revenue                0
dtype: int64

```

```

In [31]: #Question6
number=dataframe.select_dtypes(include=['int64', 'float64'])
print("Mean of Data")
number.mean()

```

Mean of Data

```
Out[31]: Administrative      2.315166
Administrative_Duration    80.818611
Informational              0.503569
Informational_Duration     34.472398
ProductRelated            31.731468
ProductRelated_Duration   1194.746220
BounceRates               0.022191
ExitRates                 0.043073
PageValues                5.889258
SpecialDay                0.061427
OperatingSystems          2.124006
Browser                   2.357097
Region                    3.147364
TrafficType               4.069586
dtype: float64
```

```
In [32]: print("Median of Data")
number.median()
```

Median of Data

```
Out[32]: Administrative      1.000000
Administrative_Duration     7.500000
Informational               0.000000
Informational_Duration      0.000000
ProductRelated             18.000000
ProductRelated_Duration    598.936905
BounceRates                0.003112
ExitRates                  0.025156
PageValues                 0.000000
SpecialDay                 0.000000
OperatingSystems           2.000000
Browser                    2.000000
Region                     3.000000
TrafficType                2.000000
dtype: float64
```

```
In [33]: print("Mode of Data")
number.mode()
```

Mode of Data

```
Out[33]:   Administrative  Administrative_Duration  Informational  Informational_Duratio
0          0          0          0.0          0          0.
```

```
In [35]: #Question7
dataframe.fillna(number.mode)
```

Out[35]:

	Administrative	Administrative_Duration	Informational	Informational_Du
0	0	0.0	0	
1	0	0.0	0	
2	0	0.0	0	
3	0	0.0	0	
4	0	0.0	0	
...	...	...	...	
12325	3	145.0	0	
12326	0	0.0	0	
12327	0	0.0	0	
12328	4	75.0	0	
12329	0	0.0	0	

12330 rows × 18 columns

In [36]:

```
#Question 8 subset 1  
februaryData = dataframe[dataframe['Month']=="Feb"]  
print(februaryData)
```

	Administrative	Administrative_Duration	Informational	\
0	0	0.0	0	
1	0	0.0	0	
2	0	0.0	0	
3	0	0.0	0	
4	0	0.0	0	
..	...	...	...	
179	0	0.0	0	
180	0	0.0	0	
181	0	0.0	0	
182	0	0.0	0	
183	0	0.0	0	

	Informational_Duration	ProductRelated	ProductRelated_Duration	\
0	0.0	1	0.000000	
1	0.0	2	64.000000	
2	0.0	1	0.000000	
3	0.0	2	2.666667	
4	0.0	10	627.500000	
..	...	...	...	
179	0.0	6	163.250000	
180	0.0	17	284.250000	
181	0.0	1	0.000000	
182	0.0	1	0.000000	
183	0.0	8	260.000000	

	BounceRates	ExitRates	PageValues	SpecialDay	Month	OperatingSystems	\
0	0.200000	0.200000	0.0	0.0	Feb	1	
1	0.000000	0.100000	0.0	0.0	Feb	2	
2	0.200000	0.200000	0.0	0.0	Feb	4	
3	0.050000	0.140000	0.0	0.0	Feb	3	
4	0.020000	0.050000	0.0	0.0	Feb	3	
..	...	...	...	...	...	...	
179	0.033333	0.033333	0.0	0.6	Feb	1	
180	0.000000	0.011765	0.0	0.0	Feb	3	
181	0.200000	0.200000	0.0	0.6	Feb	4	
182	0.200000	0.200000	0.0	0.0	Feb	2	
183	0.008333	0.025000	0.0	0.8	Feb	3	

	Browser	Region	TrafficType	VisitorType	Weekend	Revenue
0	1	1	1	Returning_Visitor	False	False
1	2	1	2	Returning_Visitor	False	False
2	1	9	3	Returning_Visitor	False	False
3	2	2	4	Returning_Visitor	False	False
4	3	1	4	Returning_Visitor	True	False
..	...	...	...	...	...	...
179	1	6	2	Returning_Visitor	False	False
180	2	6	2	Returning_Visitor	True	False
181	2	1	3	Returning_Visitor	False	False
182	4	1	1	Returning_Visitor	False	False
183	2	1	3	Returning_Visitor	True	False

[184 rows x 18 columns]

```
In [37]: #Question 8 subset 2  
set2 = dataframe[dataframe['ExitRates']>0.1]  
set2 = set2[set2['OperatingSystems']==3]  
print(set2)
```

	Administrative	Administrative_Duration	Informational	\
3	0	0.0	0	
21	0	0.0	0	
47	0	0.0	0	
56	0	0.0	0	
69	0	0.0	0	
...	...	...	...	
11775	0	0.0	0	
11993	0	0.0	0	
12090	0	0.0	0	
12174	0	0.0	0	
12197	0	0.0	0	

	Informational_Duration	ProductRelated	ProductRelated_Duration	\
3	0.0	2	2.666667	
21	0.0	2	0.000000	
47	0.0	3	80.000000	
56	0.0	1	0.000000	
69	0.0	1	0.000000	
...	...	...	...	
11775	0.0	9	38.750000	
11993	0.0	1	0.000000	
12090	0.0	5	43.500000	
12174	0.0	1	0.000000	
12197	0.0	3	0.000000	

	BounceRates	ExitRates	PageValues	SpecialDay	Month	OperatingSystems
3	0.050000	0.140000	0.0	0.0	Feb	3
21	0.200000	0.200000	0.0	0.0	Feb	3
47	0.066667	0.133333	0.0	0.2	Feb	3
56	0.200000	0.200000	0.0	0.6	Feb	3
69	0.200000	0.200000	0.0	0.2	Feb	3
...	...	...	...	...	...	...
11775	0.120000	0.120833	0.0	0.0	Nov	3
11993	0.200000	0.200000	0.0	0.0	Dec	3
12090	0.140000	0.180000	0.0	0.0	Nov	3
12174	0.200000	0.200000	0.0	0.0	Dec	3
12197	0.200000	0.200000	0.0	0.0	Nov	3

	Browser	Region	TrafficType	VisitorType	Weekend	Revenue
3	2	2	4	Returning_Visitor	False	False
21	3	1	3	Returning_Visitor	False	False
47	2	1	3	Returning_Visitor	False	False
56	2	3	3	Returning_Visitor	False	False
69	2	3	3	Returning_Visitor	False	False
...	...	...	...	...	...	...
11775	2	1	13	Returning_Visitor	False	False
11993	2	1	1	Returning_Visitor	False	False
12090	3	1	3	Returning_Visitor	False	False
12174	2	9	1	Returning_Visitor	True	False
12197	2	1	10	Returning_Visitor	False	False

[340 rows x 18 columns]

```
In [38]: #Question 8 subset 3  
TrueRev = dataframe[dataframe['Revenue']==True]  
print(TrueRev)
```

	Administrative	Administrative_Duration	Informational	\
65	3	87.833333	0	
76	10	1005.666667	0	
101	4	61.000000	0	
188	9	111.500000	1	
196	2	56.000000	1	
...	...	...	...	
12272	6	133.466667	0	
12276	7	139.575000	0	
12311	1	0.000000	2	
12312	7	150.357143	1	
12313	3	16.000000	3	

	Informational_Duration	ProductRelated	ProductRelated_Duration	\
65	0.00	27	798.333333	
76	0.00	36	2111.341667	
101	0.00	19	607.000000	
188	48.50	49	1868.819697	
196	144.00	67	2563.783333	
...	...	...	...	
12272	0.00	44	2664.445833	
12276	0.00	30	986.500000	
12311	211.25	144	4627.489571	
12312	9.00	221	11431.001240	
12313	86.00	15	2773.500000	

	BounceRates	ExitRates	PageValues	SpecialDay	Month	OperatingSystems	\
65	0.000000	0.012644	22.916036	0.8	Feb		2
76	0.004348	0.014493	11.439412	0.0	Feb		2
101	0.000000	0.026984	17.535959	1.0	Feb		1
188	0.000000	0.020709	1.706015	0.0	Mar		2
196	0.000000	0.005797	19.342650	0.0	Mar		2
...	...	...	...	...	...		...
12272	0.002041	0.010884	97.860836	0.0	Nov		2
12276	0.000000	0.011429	36.392861	0.0	Dec		2
12311	0.001361	0.020664	0.000000	0.0	Nov		2
12312	0.011149	0.021904	1.582473	0.0	Nov		2
12313	0.000000	0.030000	78.811725	0.0	Dec		2

	Browser	Region	TrafficType	VisitorType	Weekend	Revenue
65	2	3	1	Returning_Visitor	False	True
76	6	1	2	Returning_Visitor	False	True
101	1	7	4	Returning_Visitor	True	True
188	2	7	2	Returning_Visitor	False	True
196	2	4	2	New_Visitor	False	True
...	...	...	...	...	...	...
12272	2	1	3	Returning_Visitor	True	True
12276	10	1	2	New_Visitor	False	True
12311	2	1	2	Returning_Visitor	False	True
12312	5	1	2	Returning_Visitor	True	True
12313	2	1	2	Returning_Visitor	False	True

[1908 rows x 18 columns]

```
In [39]: #Question 9
y = dataframe['Revenue']
x = dataframe.drop('Revenue',axis=1)
x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.2)
print(x_train.head)
print("-----")
print(x_test.head)
```

```

<bound method NDFrame.head of
formational \
4429      0      0.0      0
1703      1     102.5      0
1041      9     122.0      1
9454      0      0.0      0
2516      0      0.0      2
...      ...      ...      ...
1966      0      0.0      0
1939      4      55.5      0
4492      0      0.0      0
6672      3      54.8      1
2182      0      0.0      0

```

```

Informational_Duration  ProductRelated  ProductRelated_Duration \
4429      0.0      3      106.500000
1703      0.0      5      97.000000
1041      87.0     50     1694.797619
9454      0.0     12     115.000000
2516      52.0     12     699.250000
...      ...      ...      ...
1966      0.0     17     588.666667
1939      0.0     33     1274.190476
4492      0.0     10     140.000000
6672      28.2     24     203.067778
2182      0.0      1      0.000000

```

```

BounceRates  ExitRates  PageValues  SpecialDay  Month  OperatingSystems \
4429      0.000000      0.033333      0.000000      0.0  May      3
1703      0.000000      0.011111      0.000000      0.0  Mar      3
1041      0.008929      0.018155      0.000000      0.0  Mar      1
9454      0.083333      0.100000      0.000000      0.0  Dec      3
2516      0.023810      0.028571      0.000000      0.0  May      1
...      ...      ...      ...      ...      ...
1966      0.011765      0.035294      0.000000      0.0  Mar      2
1939      0.005714      0.009286      0.000000      0.0  Mar      2
4492      0.040000      0.070000      0.000000      0.8  May      1
6672      0.000000      0.008000      44.359726      0.0  June     2
2182      0.200000      0.200000      0.000000      0.0  May      3

```

```

Browser  Region  TrafficType  VisitorType  Weekend
4429      2      2      13  Returning_Visitor  False
1703      2      1      9    New_Visitor      True
1041      1      1      3  Returning_Visitor  True
9454      2      8      1  Returning_Visitor  False
2516      1      3      3  Returning_Visitor  False
...      ...      ...      ...      ...
1966      4      4      1  Returning_Visitor  False
1939      2      6      2  Returning_Visitor  True
4492      8      3      1  Returning_Visitor  False
6672      2      1      4  Returning_Visitor  False
2182      2      3      1  Returning_Visitor  False

```

[9864 rows x 17 columns]>

```

-----
<bound method NDFrame.head of
nformational \
11409      0      0.000000      3
3128      0      0.000000      0
2752      5     138.500000      0
12311     1      0.000000      2
491       0      0.000000      0
...      ...      ...      ...
8251      0      0.000000      0
10077     4     208.888889      4
4801      0      0.000000      0
7556      8     203.600000      1
6284      3      48.400000      0

```

```

      Informational_Duration  ProductRelated  ProductRelated_Duration \
11409      82.50      9      143.375000
3128      0.00      5      17.000000
2752      0.00     11     578.333333
12311     211.25     144    4627.489571
491       0.00      1     10.000000
...      ...      ...      ...
8251      0.00      8     642.000000
10077     134.00     37    1335.809127
4801      0.00     29    1131.315476
7556      0.00     42    1052.400000
6284      0.00    137    3004.628109

```

```

      BounceRates  ExitRates  PageValues  SpecialDay  Month  OperatingSystems
\
11409      0.016667  0.055556  0.000000      0.0  Dec      3
3128      0.000000  0.040000  0.000000      0.0  May      4
2752      0.050000  0.072917  0.000000      0.0  May      1
12311     0.001361  0.020664  0.000000      0.0  Nov      2
491       0.000000  0.100000  0.000000      0.0  Mar      3
...      ...      ...      ...      ...      ...
8251      0.000000  0.028571  109.121156      0.0  Dec      2
10077     0.024242  0.038461  6.062330      0.0  Dec      1
4801      0.021429  0.046071  0.000000      0.0  May      2
7556      0.000000  0.004545  37.357258      0.0  Nov      2
6284      0.002381  0.011750  0.000000      0.0  Oct      2

```

```

      Browser  Region  TrafficType  VisitorType  Weekend
11409      2      6      2  Returning_Visitor  False
3128      2      3      1  Returning_Visitor  False
2752      1      1      4  Returning_Visitor  False
12311     2      1      2  Returning_Visitor  False
491       2      3      9  Returning_Visitor  True
...      ...      ...      ...      ...
8251      2      3      2      New_Visitor  False
10077     1      6      1  Returning_Visitor  False
4801      2      2      6  Returning_Visitor  True
7556      2      5      2  Returning_Visitor  False
6284      4      9      2  Returning_Visitor  False

```

[2466 rows x 17 columns]>

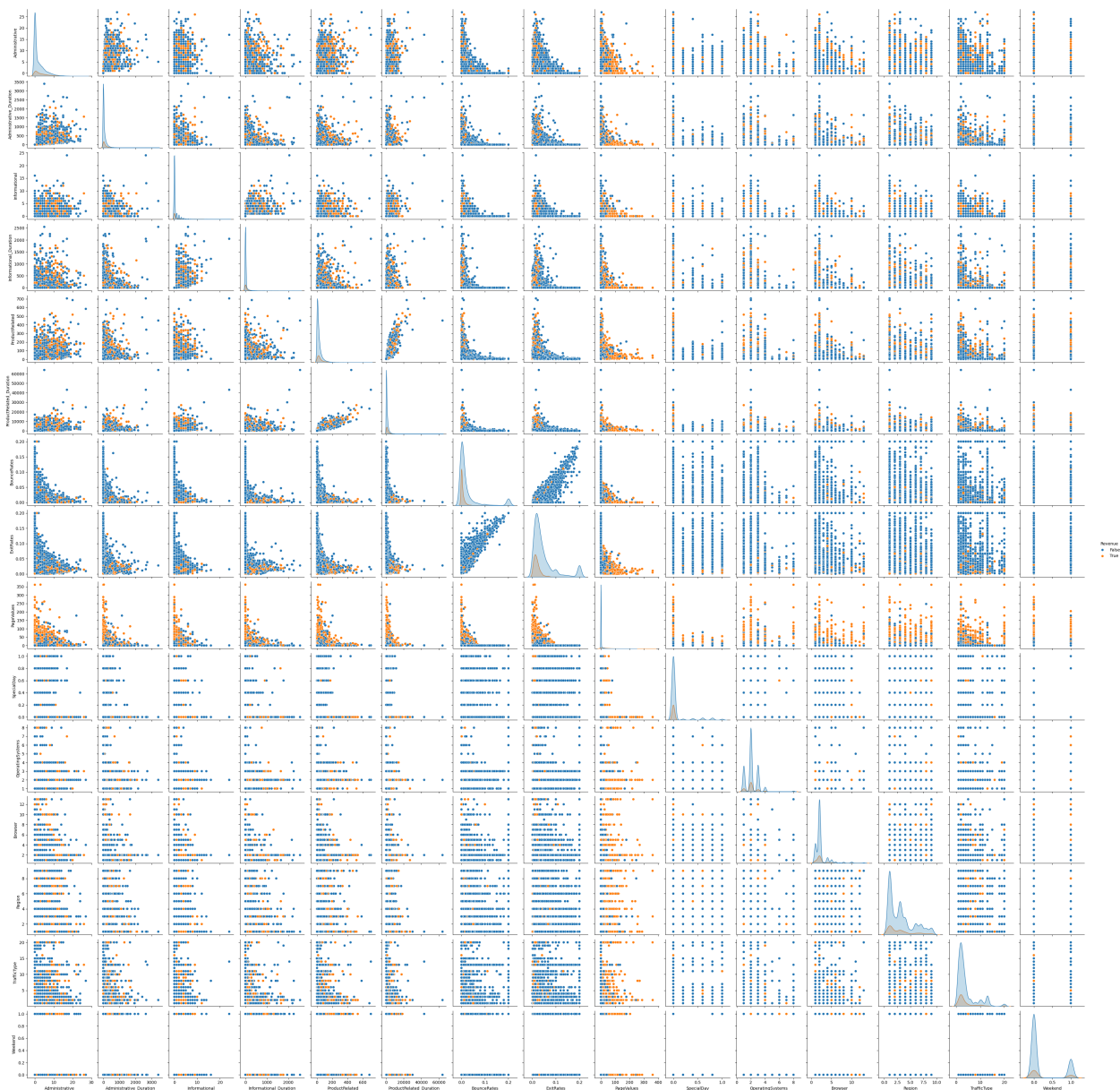
```
In [41]: #Question 10
dataframe.describe()
```

```
Out[41]:
```

	Administrative	Administrative_Duration	Informational	Informational_Du
count	12330.000000	12330.000000	12330.000000	12330.0
mean	2.315166	80.818611	0.503569	34.4
std	3.321784	176.779107	1.270156	140.7
min	0.000000	0.000000	0.000000	0.0
25%	0.000000	0.000000	0.000000	0.0
50%	1.000000	7.500000	0.000000	0.0
75%	4.000000	93.256250	0.000000	0.0
max	27.000000	3398.750000	24.000000	2549.3

```
In [44]: sns.pairplot(dataframe,hue='Revenue')
```

```
Out[44]: <seaborn.axisgrid.PairGrid at 0x1a53d8c6390>
```



In []: